

LAKSHAY BATRA

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EDUCATION

Thapar Institute of Engineering and Technology (TIET), Patiala
B.Tech in Computer Engineering
Scholars Rosary Sr. Sec. School, Rohtak
CBSE (Class XII)

2023 – 2027
CGPA: 7.58/10
Graduated Mar 2023
Percentage: 92%

EXPERIENCE

Research Intern

Jun 2025 – Jul 2025
Patiala, India

Thapar Institute of Engineering and Technology

- Engineered a **modular asynchronous ingestion and preprocessing pipeline** for large-scale physiological signal datasets, decoupling feature extraction from model training.
- Extracted **RR intervals and 8+ respiratory features** from ECG-derived respiration signals, and ran exploratory analysis on **PhysioNet data** to validate signal quality before training.
- Deployed a novel **SVM + Random Forest + LSTM ensemble** for apnea event prediction, reaching **95% test accuracy** and cutting **false positives by 15%**.
- Built reusable **ETL-style processing modules** for the **PTB-XL and MIT-BIH datasets**, eliminating **20+ hours/week** of manual preprocessing.

PROJECTS

IntelliFlow – Workflow Automation Platform

React.js, Node.js, MongoDB, LangGraph, FastAPI

- Architected an **agentic workflow platform** converting client requests into structured projects through **AI classification, dynamic task breakdown, employee matching, and approval workflows**.
- Engineered a **React + TypeScript** frontend and **Node.js + Express** REST API with **MongoDB/Mongoose**, supporting client, employee, and manager portals across the project lifecycle.
- Built a **Python FastAPI AI microservice** using **LangGraph + Groq LLMs** for request classification and workflow generation with **Pydantic validation, timeouts, and fallbacks**.
- Implemented **employee matching** using skills, workload, availability, and department scoring, alongside **JWT, bcrypt, OTP-based 2FA, rate limiting, and security middleware**.

Keystroke Authentication System

Python, Scikit-learn, XGBoost, SVM

- Developed a **behavioral biometric authentication system** analyzing **31 keystroke-timing features** to authenticate users based on unique typing patterns.
- Benchmarked **XGBoost, SVM, Random Forest, and ensemble models** for **binary user authentication**, achieving **94% accuracy** in distinguishing genuine users from impostors.
- Engineered an **end-to-end ML pipeline** processing **400 typing samples per user**, incorporating **feature engineering, standardization, and independent training/testing splits**.
- Applied **hyperparameter tuning and threshold optimization** to improve model discrimination between **genuine and impostor attempts**.

TECHNICAL SKILLS

Languages: C++, Python, JavaScript, TypeScript, SQL, HTML5, CSS3
Web & Backend: React.js, Node.js, Express.js, Tailwind CSS, REST APIs, JWT
Databases: MongoDB, ChromaDB, MySQL
Machine Learning: Scikit-learn, XGBoost, Pandas, NumPy, OpenCV
Generative AI: Prompt Engineering, LangChain, LangGraph, RAG, MCP
Tools: Git, GitHub, Jira, Postman
Core Concepts: DSA, OOP, DBMS, Operating Systems

LEADERSHIP

Entrepreneurship Development Cell (EDC), TIET

- Coordinated **event logistics, operations, and execution** for **E-Summit**, collaborating with organizing teams to manage on-ground activities for **500+ participants**.
- Led **business modeling and pitching workshops** for **100+ students**, facilitating sessions on **startup ideation, business strategy, and pitching**.

CERTIFICATIONS

Machine Learning Specialization – Stanford Online and DeepLearning.AI
Microsoft Azure Fundamentals (AZ-900) – Microsoft

2025
2025